

OP-Follower: Onboard Vision-based Obstacle-Aware Predictive Person Following for Quadraped Robots via IBVS-NMPC Control

Junwei Ge¹, Xinlong Miao¹, Jinya Su¹, Shihua Li¹, *IEEE Fellow*, Wen-Hua Chen², *IEEE Fellow*

Abstract—This paper presents OP-Follower, a fully onboard, pure vision-based, obstacle-aware person-following framework for quadraped robots. Unlike existing approaches that rely on LiDAR, infrastructure, or offboard computation, OP-Follower operates exclusively on visual inputs and runs entirely on a Unitree Go2. OP-Follower bridges the gap between low-frequency, computation delay-prone visual perception and safety-aware high-rate control by integrating YOLO-World-based open-vocabulary person detection, BoxMOT multi-object tracking, and Kalman filter-based delay-aware prediction within a unified planning-control architecture that tightly couples image-based visual servoing with nonlinear model predictive control. A decoupled artificial potential field-based visual obstacle avoidance strategy is further employed to ensure collision-aware navigation without interfering with target tracking objectives. Comprehensive evaluations in high-fidelity Gazebo simulations and real-world indoor and outdoor environments, including long-distance following, dynamic pursuit-evasion interactions, and multi-person scenarios, demonstrate robust real-time performance with fully onboard processing at a 50 Hz control rate. Compared with a non-predictive baseline, OP-Follower reduces pixel-space and distance tracking errors by up to 57.2% and 50.5%, achieving representative root-mean-square errors of approximately 13–16 pixels and 0.09–0.11 m under sharp turning motions, while maintaining stable outdoor performance including a continuous 1.4 km trial. GitHub page with demo videos is available at <https://op-follower.github.io/>.

Index Terms—Robot Person Following, Detection and Tracking, Image-Based Visual Servo (IBVS), Model Predictive Control, Quadraped Robot

I. INTRODUCTION

Vision-based robot control is a key enabling technology for autonomous robots, with wide applications in agriculture, mapping, and exploration [1]–[4]. Among vision-driven capabilities, person-following represents a fundamental functionality for robots operating in human-robot interaction, security patrol, and service robotics [4]–[7]. Compared with wheeled

platforms [7], quadraped robots offer superior terrain adaptability and mobility, enabling them to accompany humans across complex environments [5], [8], [9]. To function effectively in such scenarios, a quadraped follower must continuously perceive and track a moving human target, predict future motion, avoid surrounding obstacles, and generate smooth and safe locomotion commands, all under the constraints of onboard computation and noisy sensory inputs [4].

Although considerable progress has been made in robot perception and control, enabling robust person following on quadraped robots remains challenging, especially when relying solely on onboard vision. Many existing approaches use additional sensing modalities such as LiDAR, UWB beacons, motion capture systems, or body-worn tags to obtain reliable distance or orientation cues [5], [10]. While effective, these solutions significantly increase system complexity and limit applicability in real-world deployments where external infrastructure or wearable devices are unavailable or undesirable. Other general navigation models [11] require offboard computation or rely heavily on pre-mapped environments, further restricting autonomy and scalability.

A pure vision-based onboard solution is more natural and broadly deployable, but introduces its own challenges. Onboard vision suffers from low update frequency, motion blur (due to the inherent attitude variations of quadraped robots in locomotion), occlusions [12], and large variations in appearance and scale [13], [14], making stable long-term target tracking difficult. Translating noisy images into actionable motion requires tight coupling between perception and control; naive geometric or decoupled controllers often fail due to delay accumulation or lack of robustness. Additionally, obstacle avoidance must run with target following while ensuring computational efficiency on embedded hardware such as the Jetson Orin NX onboard Unitree Go2 in this study. Achieving accurate perception, predictive control, and safe navigation simultaneously under limited onboard resources of a quadraped robot remains an open research challenge.

To address these challenges, we propose **OP-Follower**, a unified fully onboard vision-based framework for real-time, obstacle-aware person following on quadraped robots. The core idea is to explicitly bridge the mismatch between low-frequency, delay-prone visual perception and the high-rate, safety-aware control required for stable human-robot interaction under onboard computational constraints. OP-Follower integrates open-vocabulary person detection, delay-aware predictive visual servo tracking [15], nonlinear model predictive control (NMPC) [10], [16], [17], and decoupled visual obstacle avoidance into a cohesive perception-control pipeline that runs

*This work was supported by National Natural Science Foundation of China under Grant 62303110, Start-Up Research Fund of Southeast University under Grant RF1028623226, the National Industrial Internet Industry-Education Integration Platform, Southeast University, Research Grants Council of the Hong Kong Special Administrative Region, China, under Grant C6078-25G, Basic Research Program of Jiangsu under Grant No. BK20253020, and Xiaomi Young Talents Program. Corresponding author: Jinya Su.

¹Junwei Ge, Xinlong Miao, Jinya Su, and Shihua Li are with School of Automation, Key Laboratory of Measurement and Control of CSE, Ministry of Education, and Institute of Intelligent Unmanned Systems, Southeast University, Nanjing 210096, China. 220242214@seu.edu.cn; 220242088@seu.edu.cn; sucas@seu.edu.cn; lsh@seu.edu.cn.

²Wen-Hua Chen is with the Research Centre for Low Altitude Economy and the Department of Aeronautical and Aviation Engineering, The Hong Kong Polytechnic University, Hong Kong, China. wenhua.chen@polyu.edu.hk

entirely on a Unitree Go2 equipped with a Jetson Orin NX 16GB. By combining modern visual detection models with Kalman filter (KF)-based motion prediction and a tightly coupled Image-Based Visual Servoing (IBVS) and NMPC architecture, the proposed framework enables stable and reliable person following in dynamic and cluttered environments, without relying on auxiliary sensors or offboard computation.

The proposed OP-Follower framework is rigorously validated through extensive high-fidelity Gazebo simulations and real-world experiments across diverse indoor and outdoor settings. To the best of our knowledge, this work presents fully onboard, pure vision-based person-following systems for quadruped robots that jointly integrates open-vocabulary perception, delay-aware predictive visual tracking, and safety-constrained predictive control without relying on auxiliary sensing devices such as LiDAR or angle sensors. The experimental results demonstrate that explicitly addressing perception latency within a predictive control framework is a key enabler for robust and scalable vision-based person following on resource-constrained quadruped platforms. To explicitly distinguish our methodological novelties from system-level integration, our main contributions are summarized as follows:

- (1) **Asynchronous Visual State Propagation:** We propose an asynchronous Kalman Filter-based mechanism [18] that bridges the perception-control bandwidth mismatch, providing continuous, control-rate predicted states under low-rate and intermittent visual perception.
- (2) **Delay-Aware Predictive Visual Servoing:** We develop an IBVS-NMPC formulation strictly built on these predicted visual states, enabling constrained, anticipatory person following that explicitly mitigates the destabilizing effects of perception latency.
- (3) **Fully Onboard System Integration and Validation:** We achieve a fully onboard, pure vision-based framework on a quadruped robot that operates reliably at a 50 Hz control rate, without relying on LiDAR, external infrastructure, or offboard computation.

Notations: **RPF:** Robot Person Following; **IBVS:** Image-Based Visual Servoing; **MPC:** Model Predictive Control; **KF:** Kalman Filter; **APF:** Artificial Potential Field.

II. RELATED WORK

A. Person Detection and Tracking

Advances in visual perception have substantially improved person detection and tracking for mobile robots. Modern deep detectors enable robust human recognition under appearance variations and cluttered environments, while open-vocabulary detection further enhances generalization across unseen scenes. Representative examples include YOLO-World [19], which shows real-time open-vocabulary detection suitable for embedded platforms. To ensure temporal consistency, tracking-by-detection pipelines [1] combined with multi-object tracking are widely adopted to maintain stable identities under occlusions and multi-person interactions [20].

Despite these advances, onboard vision for quadruped robots still suffers from low update rates, motion blur induced

by locomotion, and intermittent detection failures. Directly using such low-frequency and noisy visual measurements in closed-loop control often leads to delayed responses and degraded tracking stability. This motivates the incorporation of delay-aware state estimation and prediction mechanisms to bridge the gap between visual perception and control.

B. Robot Person Following

Robot person following (RPF), as summarized in Table I, has been extensively studied across wheeled, aerial, and legged platforms. Many existing approaches rely on auxiliary sensing modalities, such as LiDAR, UWB beacons, wearable tags, or external localization systems, to obtain reliable relative pose information [5]. While effective, these solutions increase system complexity and limit deployability in real-world environments. Pure vision-based RPF systems provide a more natural and infrastructure-free alternative, but often adopt reactive control [21] strategies that are sensitive to perception delays and prone to oscillations or target loss.

Recent learning-based and foundation-model-driven approaches demonstrate strong generalization capabilities for visual tracking and navigation [4], [8], [11]. However, such methods typically require heavy computation, offboard processing, or lack explicit safety and constraint handling at the control level. Consequently, an open gap remains in developing a *fully onboard, pure vision-based* RPF system that jointly accounts for perception delays, predictive motion, and collision avoidance.

C. MPC and Visual Servoing for Vision-Driven Robots

Visual servoing provides a principled framework for regulating robot motion directly in image space [2]. Image-Based Visual Servoing (IBVS) is particularly attractive due to its robustness to calibration errors and its direct use of visual feedback [15], [22]. However, classical IBVS is inherently reactive and does not explicitly consider constraints such as field-of-view limits, actuator bounds, or obstacle avoidance.

To address these limitations, predictive control techniques have been integrated with visual servoing, leading to MPC-based formulations that explicitly handle constraints and improve anticipatory behavior [23], [24]. These studies demonstrate the potential of combining IBVS with MPC for safe and smooth vision-driven control [15]. Nevertheless, deploying such predictive visual servoing frameworks in real-world person-following tasks remains challenging due to low-rate perception, target motion uncertainty, and potential conflicts between tracking and obstacle avoidance. This motivates our unified IBVS-NMPC formulation with delay-aware prediction and decoupled visual obstacle avoidance.

III. PRELIMINARIES AND PROBLEM FORMULATION

In this work, we consider a quadruped robot equipped with a forward-facing stereo camera rigidly mounted on its head. The robot coordinate frame is denoted by $\mathbf{P}_R$, and the camera coordinate frame by $\mathbf{P}_C$, as shown in Fig. 1. The stereo camera provides both pixel coordinates and depth information for the

TABLE I: Comparisons of representative robot person-following methods and the remaining gap. **Blue** indicates our proposed method.

Ref.	Year	N.G.M.	Obs.	Pre.	OnB.	Aux.
[25]	2025	×	✓	✓	✓	—
[5]	2025	✓	✓	×	✓	Angle Sensor
[26]	2025	✓	✓	×	×	—
[4]	2024	✓	×	✓	✓	—
[6]	2022	✓	✓	×	✓	UWB
Ours	2026	✓	✓	✓	✓	—

Open Gap: Existing methods typically address only subsets of these requirements. In contrast, OP-Follower jointly considers onboard execution, predictive tracking under perception delay, and collision avoidance within a unified visual control framework.

N.G.M. the tracking process relies on no global map or not; Obs. whether obstacle avoidance is supported; Pre. whether target-motion prediction is considered; OnB. whether the method runs fully onboard; Aux. What auxiliary methods were used.

target person and surrounding obstacles, forming the primary sensory input for our fully vision-based RPF framework.

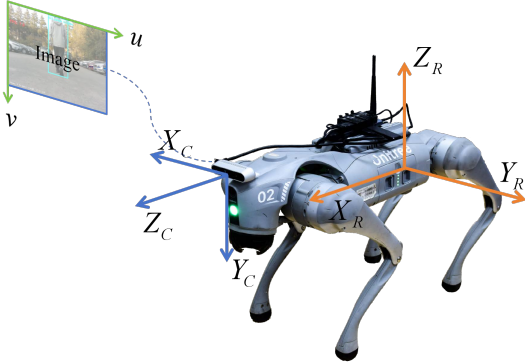


Fig. 1: Coordinates of the quadruped robot and the camera.

A. Camera Projection and Image Feature

For a 3-D point with coordinates $\mathbf{X} = (X_c, Y_c, Z_c)$ in the camera frame, its perspective projection onto the image plane yields a 2-D feature $\mathbf{s} = (x, y)$, which is related through:

$$\begin{cases} x = \frac{X_c}{Z_c} = \frac{u - c_x}{f_x} \\ y = \frac{Y_c}{Z_c} = \frac{v - c_y}{f_y} \end{cases}, \quad (1)$$

where $\mathbf{m} = (u, v)$ denotes the pixel coordinates, (f_x, f_y) is the focal length, and (c_x, c_y) is the principal point of the camera. These relations allow the conversion of visual observations into normalized image-space features, which serve as states for visual servoing and predictive control.

B. Interaction Matrix and Pixel-space Motion Model

Following the classical visual servoing theory [15], [22], the temporal evolution of the image feature vector is governed by:

$$\dot{\mathbf{s}} = \mathbf{L}_s \mathbf{v}_c, \quad (2)$$

where $\mathbf{v}_c = [v_x^c, v_y^c, v_z^c, \omega_x^c, \omega_y^c, \omega_z^c]^T$ is the camera spatial velocity, and $\mathbf{L}_s$ is the interaction matrix, given by

$$\mathbf{L}_s = \begin{bmatrix} -\frac{1}{Z_c} & 0 & \frac{x}{Z_c} & xy & -(1+x^2) & y \\ 0 & -\frac{1}{Z_c} & \frac{y}{Z_c} & 1+y^2 & -xy & -x \end{bmatrix}. \quad (3)$$

Transforming (2) into pixel coordinates yields:

$$\dot{\mathbf{m}} = \mathbf{L}_m \mathbf{v}_c, \quad (4)$$

where pixel-domain interaction matrix $\mathbf{L}_m$ is given in (5). This model provides a differential relationship between camera motion and pixel-space feature evolution, forming the basis of our IBVS–NMPC predictive controller in Section IV-D.

C. Problem Formulation

The goal of RPF is to regulate the robot motion such that:

- (1) the target person remains near the center of the image plane (i.e., image-space regulation);
- (2) the robot maintains a user-defined socially appropriate following distance $Z_c \in [Z_{\min}, Z_{\max}]$;
- (3) the robot avoids collisions with obstacles perceived through vision;
- (4) the control inputs remain smooth and within feasible limits for quadruped locomotion.

However, the tracking process is also subject to:

- (a) person motion uncertainty and abrupt velocity changes;
- (b) noisy and low-frequency visual perception caused by the computational burden of state-of-the-art (SOTA) detection and tracking algorithms;
- (c) partial or temporary object occlusions;
- (d) dynamic obstacles and cluttered environments;
- (e) computational constraints of the onboard Jetson Orin NX.

The goal is to compute a feasible control policy that ensures stable person following and reliable obstacle avoidance under the uncertainties of real-world visual perception, using only onboard vision without any auxiliary sensors.

IV. OP-FOLLOWER FRAMEWORK

We decompose the RPF task into four tightly coupled subtasks, as illustrated in Fig. 2: Visual Target Detection & Tracking (Section IV-A), Delay-Aware Prediction (Section IV-B), Visual Obstacle Avoidance (Section IV-C), and IBVS–NMPC Tracking Control (Section IV-D). First, the onboard stereo camera provides pixel coordinates and depth information for the target person and surrounding obstacles. A KF-based state estimator compensates for low-frequency perception updates by predicting person and obstacle states, enabling high-rate tracking and obstacle avoidance control. The IBVS–NMPC module exploits these pixel-space features and the prediction model in Section III to achieve robust and natural person following, while a visual obstacle avoidance module uses the same visual inputs to generate a decoupled lateral motion command that does not interfere with the following behavior. Together, these components form a complete, pure vision-based RPF framework that explicitly accounts for person motion uncertainty and dynamic obstacles in real-world environments. The core modules of OP-Follower are detailed below.

$$\mathbf{L}_m = \begin{bmatrix} -\frac{f_x}{Z_c} & 0 & \frac{u-c_x}{Z_c} & \frac{(u-c_x)(v-c_y)}{f_y} & -\left[f_x + \frac{(u-c_x)^2}{f_x}\right] & \frac{f_x(v-c_y)}{f_x} \\ 0 & -\frac{f_y}{Z_c} & \frac{v-c_y}{Z_c} & f_y + \frac{(v-c_y)^2}{f_y} & -\frac{(u-c_x)(v-c_y)}{f_x} & -\frac{f_y(u-c_x)}{f_x} \end{bmatrix}. \quad (5)$$

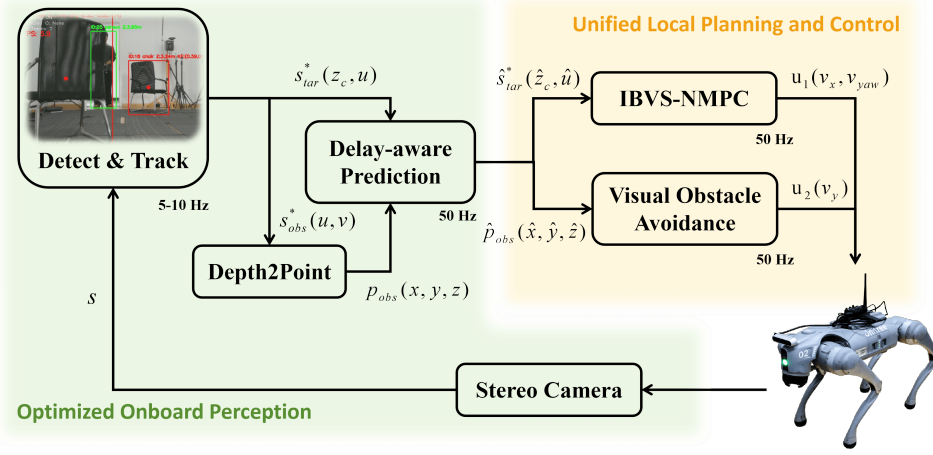


Fig. 2: **Overall architecture of the proposed OP-Follower framework**, seamlessly integrating YOLO-World detection, BoxMOT tracking, KF prediction, IBVS-NMPC and APF obstacle avoidance.

A. Visual Detection & Tracking

Accurate and robust person and obstacle detection forms the foundation of the RPF pipeline. In our framework, this is achieved through a multi-stage perception module that combines SOTA object detection with efficient multi-object tracking. We employ YOLO-World [19] as the core detection engine due to its open-vocabulary capability and favorable trade-off between inference speed and accuracy, which is essential for real-time execution on embedded platforms. The model is pre-trained on large-scale datasets and adapted to reliably detect the *person* class in cluttered environments, outputting bounding boxes in pixel coordinates (u, v, w, h) with associated confidence scores.

To ensure consistent identity assignment across frames, especially under occlusions or in multi-person scenarios, we integrate the BoxMOT [20] multi-object tracking algorithm. This detection-tracking pipeline provides, for each frame, the target person’s pixel coordinates (u, v) , stereo-estimated depth Z_c , and a stable unique ID. The perception front-end further serves a dual purpose by naturally coupling tracking with obstacle detection. During initialization, the operator designates the target person (e.g., via an initial bounding-box selection), assigning them a unique locked *track_id*. The depth of the target is robustly extracted by applying a RANSAC regressor to the stereo depth points bounded by the 2D detection box, filtering out background noise. Crucially, all valid visual detections whose tracking IDs do not match the designated target *track_id*—including non-target pedestrians and prompted static objects—are automatically classified as dynamic or static obstacles. Their pixel coordinates and

RANSAC-filtered depths are continuously forwarded to the Visual Obstacle Avoidance module (detailed in Section IV-C), enabling real-time collision avoidance. This unified design allows a single visual pipeline to support both predictive tracking and safety without redundant computational overhead. Detailed hyperparameter settings and deployment configurations for these modules are provided in Section V-A.

B. Delay-Aware Prediction

In real-world RPF scenarios, the visual detection and tracking module typically operates at a relatively low update frequency (e.g., 5–10 Hz) due to camera limitations and the computational load of the recognition network. Directly using such low-rate measurements for control introduces non-negligible delays and can lead to unstable or jerky behavior, especially when the person moves abruptly [27].

To mitigate this perception-control bandwidth mismatch, we employ a KF as a state estimator to provide high-frequency, delay-aware predictions of the target’s motion. Considering that the RPF task mainly focuses on the x-coordinate, the pixel y-coordinate is omitted, and therefore we define the KF state as

$$\mathbf{x}^{\text{KF}} = [u \quad Z_c \quad \dot{u} \quad \dot{Z}_c]^T, \quad (6)$$

where u and Z_c denote the pixel x-coordinate and depth of the target in the camera frame, and $(\dot{u}, \dot{Z}_c)$ approximate their pixel-space and depth-space velocities. A constant-velocity model is adopted for the prediction steps in the absence of low-frequency measurements:

$$\mathbf{x}_{k+1}^{\text{KF}} = \mathbf{A} \mathbf{x}_k^{\text{KF}} + \mathbf{w}_k, \quad (7)$$

where $\mathbf{A}$ is the state transition matrix derived from the constant-velocity assumption and $\mathbf{w}_k$ is process noise. The measurement vector is given by the visual detections

$$\mathbf{y}_k = [u_k \quad Z_{c,k}]^T = \mathbf{H} \mathbf{x}_k^{\text{KF}} + \mathbf{v}_k, \quad (8)$$

where $\mathbf{H}$ denotes the measurement matrix and $\mathbf{v}_k$ is measurement noise.

While higher-order kinematic models, such as Constant Acceleration (CA), or data-driven predictors may better represent certain pedestrian maneuvers, increased model order does not necessarily improve short-horizon prediction under noisy, low-rate observations. Recent studies on pedestrian motion prediction and crowd navigation show that simple Constant Velocity (CV) models remain competitive under such conditions [28], [29]. In our visual pipeline, the horizontal bounding-box coordinate and stereo depth are affected by body articulation, posture variation, and intermittent perception noise. Estimating acceleration from these 5–10 Hz observations increases sensitivity to short-term measurement fluctuations. The CV model therefore provides a suitable first-order motion prior for the sensing conditions and prediction horizon considered in this work, balancing measurement smoothing and responsiveness to macroscopic target motion.

In this study, low-rate visual detection (5–10 Hz) is fused as measurements to correct the state, while the KF runs at the rate (50 Hz) to propagate the state forward in the absence of new measurements. It should be explicitly noted that the term “delay-aware” in this framework does not refer to a dedicated out-of-sequence measurement treatment or an explicit timestamp-based latency compensation scheme. Rather, it emphasizes that the proposed KF-based module is designed to account for the practical bandwidth mismatch between low-rate visual perception and high-rate control execution. As illustrated in Fig. 3, this approach effectively compensates for sampling mismatch and intermittent visual failures, providing continuous and smooth estimates of (u, Z_c) and their velocities. These predictions serve as key inputs for both the IBVS–NMPC tracking controller and the high-frequency obstacle avoidance module. Similarly, for detected obstacles, we adopt a similar method for their tracking and delay-aware prediction. This design effectively decouples perception update rates from control execution rates, allowing the NMPC to operate on a consistent high-frequency state estimate despite asynchronous visual measurements.

C. Visual Obstacle Avoidance

Rather than using APF as a standalone planner, we employ it as a lateral corrective mechanism (tailored to the omnidirectional motion characteristics of quadruped robots) explicitly constrained to avoid interference with the visual servoing objective. Classic local obstacle avoidance methods often rely on a 3-D occupancy or distance map combined with search-based [30], sampling-based [31], or optimization-based [3] planning. However, these map-based planners are conceptually separated from the visual servoing controller and may even conflict with it. Visual servoing can be viewed as a

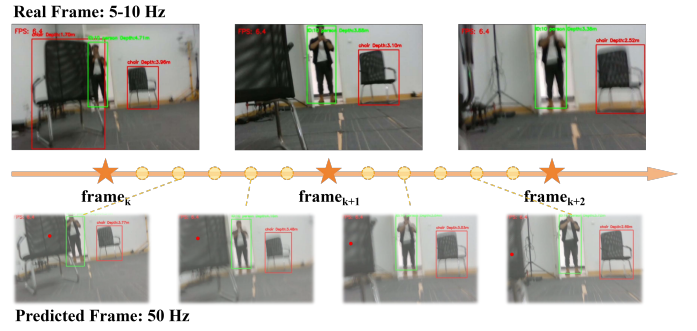


Fig. 3: KF-based delay-aware prediction of the target’s motion state.

local controller that keeps the target centered, while a separate local planner may generate commands that prioritize obstacle clearance, potentially pulling the robot away from the target.

As illustrated in Fig. 4, when the target and an obstacle appear simultaneously on the left of the robot, a conventional local planner may suggest moving right to avoid the obstacle, whereas the visual servoing module may command a leftward motion to keep the target centered in the image. Without careful coordination, this conflict can lead to target loss or even collisions.

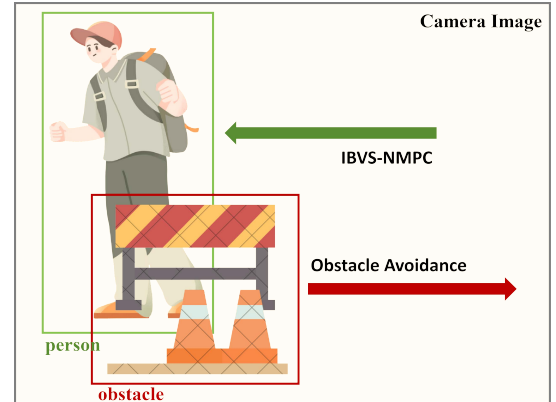


Fig. 4: Schematic of potential conflict between traditional local obstacle avoidance and visual servoing.

To resolve this issue under the visual servoing architecture, we upgrade the traditional position-based APF to a Velocity-Obstacle (VO)-aware Dynamic APF. While a classical APF effectively maintains spatial clearance, it relies solely on relative Euclidean distance, rendering it insufficient for highly dynamic environments where fast-approaching obstacles necessitate anticipatory evasive maneuvers. To improve safety against dynamic threats without interfering with the visual servoing objectives, the repulsive force formulation is restructured to scale directly with the obstacle’s approaching kinetics. Let $\mathbf{p}_{rob}$ and $\mathbf{p}_{obs,i}$ denote the Cartesian positions of the robot and the i -th obstacle, with their relative Euclidean distance defined as $\rho_i = \|\mathbf{p}_{rob} - \mathbf{p}_{obs,i}\|$. The radial unit vector pointing from the obstacle to the robot is $\mathbf{n}_{r,i} = (\mathbf{p}_{rob} - \mathbf{p}_{obs,i}) / \rho_i$. Crucially, we define the relative approach velocity as $v_{app,i} = -\dot{\rho}_i$,

where a positive value strictly indicates approaching kinetics.

The complete dynamic repulsive force $\mathbf{F}_{rep,i}^{total}$ on the robot from the i -th obstacle within the influence radius ρ_0 is formulated as:

$$\mathbf{F}_{rep,i}^{total} = \underbrace{k_{rep} \left(\frac{1}{\rho_i} - \frac{1}{\rho_0} \right) \frac{1}{\rho_i^2} (1 + \lambda \max(0, v_{app,i})) \mathbf{n}_{r,i}}_{\text{Radial Dynamic Repulsion}} + \underbrace{k_v \frac{\max(0, v_{app,i})}{\rho_i} \mathbf{n}_{t,i}}_{\text{Tangential Evasive Repulsion}} \quad (9)$$

where k_{rep} is the position repulsion gain, λ is the approach-velocity penalty coefficient, and k_v is the tangential evasive steering gain. The vector $\mathbf{n}_{t,i}$ represents the tangential unit normal ($\mathbf{n}_{t,i} \perp \mathbf{n}_{r,i}$), optimally oriented to steer the robot's velocity vector outside the anticipated collision cone.

As in (9), this formulation tightly couples spatial and kinematic semantics. The Radial Dynamic Repulsion introduces a velocity penalty factor $(1 + \lambda \max(0, v_{app,i}))$ to ensure that high-speed approaching targets generate a scaled, early-stage repulsion well before critical spatial proximity thresholds are breached. Concurrently, guided by classical Velocity Obstacle (VO) principles [32], the *Tangential Evasive Repulsion* acts as a proactive lateral steering mechanism. Rather than merely decelerating the robot—which could induce tracking failure—this tangential force actively drives the robot to execute fluid side-stepping maneuvers.

The total repulsive force acting on the robot is the vector sum of the dynamic repulsive forces from all N detected obstacles within the influence radius:

$$\mathbf{F}_{rep}^{total} = \sum_{i=1}^N \mathbf{F}_{rep,i}^{total} \quad (10)$$

To translate this physical force into actionable locomotion commands, the total repulsive vector $\mathbf{F}_{rep}^{total}$ is decomposed and projected onto the robot's body-frame forward (X_R) and lateral (Y_R) axes, yielding vertical and lateral velocity commands v_x^{rep} , v_y^{rep} .

We further introduce a weighting function to blend the attractive (following) and repulsive (avoidance) commands:

$$w = \frac{1}{1 + e^{k(\rho_{min} - \rho_0)}}, \quad (11)$$

where k controls the sharpness of the transition, and ρ_{min} is the distance to the closest obstacle. To ensure a consistent command interface for the quadruped's locomotion controller, we formulate the attractive (tracking) command and the repulsive (avoidance) command as planar body-frame velocity vectors, i.e.,

$$\mathbf{v}_{att} = [v_x^{att}, 0, \omega_z^{att}]^T, \quad \mathbf{v}_{rep} = [v_x^{rep}, v_y^{rep}, 0]^T, \quad (12)$$

where $\mathbf{v}_{att}$ and $\mathbf{v}_{rep} \in \mathbb{R}^3$, v_x^{att} and ω_z^{att} are the forward and yaw velocity commands generated by the IBVS-NMPC module (detailed in Section IV-D), and v_y^{rep} is the lateral obstacle

avoidance velocity derived from the projection of $\mathbf{F}_{rep}^{total}$. The final executed command vector $\mathbf{v}_{cmd} \in \mathbb{R}^3$ is then blended as:

$$\mathbf{v}_{cmd} = (1 - w)\mathbf{v}_{att} + w\mathbf{v}_{rep}. \quad (13)$$

This unified formulation enables flexible and smooth navigation while preserving the dominance of tracking behavior when obstacles are far away and gradually prioritizing avoidance as obstacles come closer.

D. IBVS-NMPC Tracking Control

Based on the pixel-space motion model in Section III, we design an IBVS-NMPC controller to achieve smooth and robust person following. As discussed in Section IV-C, the visual servoing module controls the forward velocity v_x^r along the robot's X_R axis and the yaw rate ω_z^r around the Z_R axis. Considering the quadruped's omnidirectional locomotion capability and the need to fuse visual servoing with visual obstacle avoidance, we simplify the interaction model to capture the dominant coupling between the target's image position and these two control inputs:

$$\begin{bmatrix} \dot{Z}_c \\ \dot{u} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{u - c_x}{Z_c} & -f_x - \frac{(u - c_x)^2}{f_x} \end{bmatrix} \begin{bmatrix} v_x^r \\ \omega_z^r \end{bmatrix}, \quad (14)$$

where Z_c is the distance between the camera and the target, and u is the horizontal pixel coordinate of the target.

We define the NMPC state and input as

$$\mathbf{x} = [Z_c, u]^T, \quad \mathbf{u} = [v_x^r, \omega_z^r]^T \equiv [v_x^{att}, \omega_z^{att}]^T. \quad (15)$$

Using a forward Euler discretization with sampling interval Δt , the discrete-time prediction model is

$$\mathbf{x}_{k+1} = \zeta(\mathbf{x}_k, \mathbf{u}_k) = \mathbf{x}_k + \Delta t f(\mathbf{x}_k, \mathbf{u}_k), \quad (16)$$

where $f(\cdot)$ is derived from (14).

Remark 1: In this work, we employ the reduced visual state $\mathbf{x} = [Z_c, u]^T$ and omit the vertical image coordinate v . The relative distance Z_c is directly obtained from stereo depth measurements, while u represents the target's horizontal alignment. For person following in quasi-planar environments, these two states capture the dominant longitudinal and lateral control objectives. Moreover, excluding v reduces the controller's sensitivity to gait-induced pitch oscillations and vertical image jitter. This simplification does not directly affect the stereo depth measurement; however, it neglects the coupling among vertical image motion, camera attitude, and terrain elevation. Consequently, the reduced model may become inaccurate on uneven terrain or steep slopes, where a full image-state or 3D relative-pose representation with attitude and terrain compensation would be required.

The stage cost of NMPC is defined to penalize deviations from the desired following configuration, excessive control effort, and rapid changes in control:

$$J(\mathbf{x}_k, \mathbf{u}_k, \mathbf{u}_{k-1|k}) = \sum_{j=0}^N \underbrace{\|\mathbf{x}_{\text{ref}} - \mathbf{x}_{k+j|k}\|_{Q_x}^2}_{\text{State cost}} + \underbrace{\|\mathbf{u}_{\text{ref}} - \mathbf{u}_{k+j|k}\|_{Q_u}^2}_{\text{Input cost}} + \underbrace{\|\mathbf{u}_{k+j|k} - \mathbf{u}_{k+j-1|k}\|_{Q_{\Delta u}}^2}_{\text{Smoothness cost}}, \quad (17)$$

where $\mathbf{x}_{k+j|k}$ denotes the predicted states at $k+j$ step based on initial states $\mathbf{x}_{k|k}$ at k^{th} step and prediction model in Eq. (16). The initial states $\mathbf{x}_{k|k}$ are obtained from the KF in Section IV-B, i.e., the filtered states or prediction states depending on the availability of observations. $\mathbf{x}_{\text{ref}}$ encodes the desired distance and image position of the target, $\mathbf{u}_{\text{ref}}$ is typically set to zero (i.e., no motion when perfectly aligned), and $Q_x, Q_u, Q_{\Delta u} \in \mathbb{R}^{2 \times 2}$ are positive definite weight matrices. The smoothness term encourages gradual changes in $\mathbf{u}$, which is important for generating natural and physically feasible quadruped motions.

To enforce social-distance and field-of-view requirements, as well as actuator limits, we impose constraints on both states and inputs:

$$Z_{\min} \leq Z_c \leq Z_{\max}, \quad u_{\min} \leq u \leq u_{\max}, \quad (18)$$

$$v_{\min} \leq v_x^r \leq v_{\max}, \quad \omega_{\min} \leq \omega_z^r \leq \omega_{\max}. \quad (19)$$

The resulting IBVS–NMPC problem is formulated as:

$$\begin{aligned} \min_{\{\mathbf{u}_{k+j|k}\}} \quad & J(\mathbf{x}_k, \mathbf{u}_k, \mathbf{u}_{k-1|k}) \\ \text{s.t.} \quad & \mathbf{x}_{k+j+1|k} = \zeta(\mathbf{x}_{k+j|k}, \mathbf{u}_{k+j|k}), \\ & j = 0, \dots, N-1, \\ & \mathbf{x}_{\min} \leq \mathbf{x}_{k+j|k} \leq \mathbf{x}_{\max}, \quad j = 0, \dots, N, \\ & \mathbf{u}_{\min} \leq \mathbf{u}_{k+j|k} \leq \mathbf{u}_{\max}, \quad j = 0, \dots, N, \end{aligned} \quad (20)$$

where N is the prediction horizon. Solving (20) in a receding horizon fashion yields the first control input $\mathbf{u}_{k|k}$, which is then applied to the quadruped robot. Combined with the delay-aware prediction and decoupled obstacle avoidance described earlier, this IBVS–NMPC formulation enables OP-Follower to achieve robust, smooth, and socially compliant person following using purely visual inputs.

V. EXPERIMENTAL VALIDATION

To evaluate the effectiveness, robustness, and real-world deployability of the proposed OP-Follower framework, we conduct extensive experiments in both high-fidelity Gazebo simulations and diverse real-world environments. All evaluations are performed on a Unitree Go2 quadruped robot equipped with an onboard NVIDIA Jetson Orin NX 16GB, where the complete perception–control pipeline runs fully onboard at real-time rates without relying on external computation or auxiliary sensing modalities. The experimental campaign is designed to systematically assess: (i) robustness of visual detection and tracking, (ii) performance of the proposed

delay-aware IBVS–NMPC controller, (iii) reliability of the decoupled visual obstacle avoidance module, and (iv) overall adaptability of the system in dynamic, multi-person, and long-range following scenarios.

A. Experimental Setup

1) *Hardware Platform*: All real-world experiments are conducted on a Unitree Go2 quadruped robot using its built-in locomotion controller. An Intel RealSense D435i stereo camera is mounted on the front of the robot, providing synchronized RGB and depth measurements for visual perception. The onboard NVIDIA Jetson Orin NX executes the complete OP-Follower pipeline, including YOLO-World detection, BoxMOT tracking, KF prediction, APF-based obstacle avoidance, and IBVS–NMPC control, without any offboard computation. The final blended high-level velocity command vector $\mathbf{v}_{\text{cmd}} = [v_x^{\text{cmd}}, v_y^{\text{cmd}}, \omega_z^{\text{cmd}}]^T$ is transmitted to the robot’s built-in locomotion controller at a fixed rate of 50 Hz.

2) *Software and Parameter Settings*: All modules are implemented within the ROS framework. YOLO-World operates at 5–10 Hz depending on scene complexity, while the KF prediction, APF module, and NMPC controller are executed at 50 Hz to match the control frequency. For the visual perception front-end, the YOLO-World detection engine utilizes the lightweight *yolov8s-worldv2* model processing RGB streams at an input resolution of 640×480 . The textual prompts are explicitly configured as “person” and typical environmental obstacles (e.g., “chair”), with a detection confidence threshold set to 0.4. BoTSORT variant in BoxMOT is drawn for target tracking, and the OSNet ReID model operates in FP16 precision to ensure real-time inference onboard the edge computer.

Regarding the control back-end, the IBVS–NMPC prediction horizon is set to $N = 100$ with a discretization step of $\delta t = 0.02$ s, aligning perfectly with the 50 Hz high-level command rate sent to the built-in locomotion controller. To ensure reproducibility, the specific weighting matrices defining the NMPC stage cost in Eq. (17) are selected as follows:

$$Q_x = \text{diag}([0.7, 0.0001]),$$

$$Q_u = \text{diag}([0.5, 10]),$$

$$Q_{\Delta u} = \text{diag}([0.2, 2.0]).$$

The reference states are set to $\mathbf{x}_{\text{ref}} = [2.0, 320]^T$, corresponding to a desired social following distance of $Z_{\text{ref}} = 2.0$ m and the horizontal image center $u_{\text{ref}}^c = 320$ px. The reference input is $\mathbf{u}_{\text{ref}} = [0, 0]^T$. (It should be noted that the horizontal image center u_{ref}^c is configured as 464 in simulations and 320 in real-world trials. This variation is attributed to the different image resolutions employed: the Gazebo environment utilizes a resolution of 928×800 , whereas the Intel RealSense D435i camera in physical experiments is configured at 640×480 to prioritize onboard processing efficiency). The resulting nonlinear optimization problem, involving two states and two control inputs, is solved using ACADOS [33], with computation times consistently below 5 ms per control cycle on the Jetson platform. The social distance constraint is defined as

$Z_c \in [0.5 \text{ m}, 3 \text{ m}]$, while the field-of-view constraint enforces the image coordinate u to remain within the central 80% region of the camera image.

B. Simulation Evaluation

The complexity of the RPF task mainly arises from the target's turning behaviors, which can be categorized into smooth and sharp turns. Smooth turns typically correspond to natural obstacle avoidance or casual walking patterns, whereas sharp turns involve abrupt directional changes and pose greater challenges for RPF systems. So we evaluate OP-Follower across a range of trajectories containing both smooth and sharp turns in a high-fidelity Gazebo environment. To ensure maximum sim-to-real transferability, this environment is explicitly built and extended upon the official *unitree_guide* SDK (https://github.com/unitreerobotics/unitree_guide). Rather than employing idealized kinematic abstractions, this setup simulates the rigid-body dynamics, strict inertia tensors, and low-level joint torque controllers of the Unitree Go 2. Consequently, our visual control framework is evaluated under more realistic conditions, including the continuous gait-induced vibrations and non-linear attitude perturbations inherent to quadruped locomotion.

The simulated scenarios include U-turns (acute directional changes), L-turns (45°) (moderate turns), circular and lemniscate paths (continuous curvature variations), as well as rectangular and double-triangular trajectories (sequential straight segments with sharp angles). This set of trajectories enables a systematic assessment of tracking accuracy and responsiveness to kinematic discontinuities under controlled conditions.

Quantitative evaluation is conducted to assess performance with respect to the core control objectives. The following metrics are adopted:

- (1) **Tracking Robustness and Stability:** Evaluated using multi-lap trajectory plots to verify consistent following behavior without target loss.
- (2) **Visual Servoing Performance:** Measured by deviations in pixel space (image u -coordinate) and depth space (following distance Z_c).

These metrics are recorded throughout all simulation trials to validate the real-time performance and robustness of the proposed control system. Detailed results corresponding to each metric are presented in the following subsections.

1) *Robust Tracking Performance Across Diverse Trajectories:* The robustness of the integrated perception-control pipeline is evaluated using a diverse set of trajectories designed to challenge different aspects of the system, including continuous-curvature paths (circular, lemniscate), sharp directional changes (U-turn, L-turn), and complex sequences of straight segments and angles (rectangular, double-triangular).

As shown in Fig. 5, OP-Follower achieves consistent and stable tracking across all scenarios. The robot successfully negotiates both smooth and sharp turns without target loss, exhibiting limited overshoot and rapid recovery following abrupt directional changes. Key performance metrics, includ-

ing tracking errors in pixel plane (image u -coordinate) and following distance (Z_c), are recorded throughout the trials.

The u -coordinate is tightly regulated around the desired setpoint at the image center, with the target consistently remaining within the central 10% of the field-of-view width. Moderate oscillations are observed in the u -coordinate, which are primarily caused by variations in the perceived bounding-box width when the target undergoes lateral reorientation during turning motions. These oscillations reflect perceptual effects induced by gait dynamics rather than control instability. The following distance Z_c remains within the predefined social bounds $[0.5\text{m}, 3\text{m}]$, demonstrating that the OP-Follower effectively enforces state constraints while maintaining accurate and stable lateral and longitudinal tracking.

2) Predictive vs. Non-Predictive Control Comparison:

To isolate and quantify the contribution of the Delay-Aware Prediction module, we compare the full OP-Follower against a non-predictive (N.Pre) baseline in which the NMPC directly adopts low-frequency visual measurements (set to 5 Hz in simulation) without state estimation or prediction. This comparison highlights the impact of perception latency on closed-loop tracking performance.

As shown in Fig. 6, the non-predictive baseline exhibits clear performance degradation, particularly during sharp turns and abrupt target motion. The delayed response caused by low-rate visual feedback leads to pronounced overshoot and oscillations in both image-space alignment and following distance, resulting in visibly less smooth robot motion.

In contrast, incorporating delay-aware prediction enables the NMPC to operate on high-rate, temporally consistent state estimates, allowing anticipatory control actions. This results in smoother trajectories, reduced oscillations, and consistently lower tracking errors, as reflected in both the time-series plots in Fig. 6 and the statistical distributions in Fig. 7. The predictive horizon further facilitates earlier turn initiation, yielding more graceful motion under rapid directional changes.

TABLE II: RMSE comparison between Full OP-Follower and the non-predictive baseline on sharp-turn scenarios including U-turn and L-turn (45°) trajectories.

RMSE	u (pixels)		Z_c (meters)	
	L-Turn	U-Turn	L-Turn	U-Turn
N.Pre OP-Follower	30.9611	27.6651	0.1888	0.1955
Full OP-Follower	13.2460	15.6693	0.0935	0.1126

Quantitative results in Table II further confirm the benefits of delay-aware prediction. For the L-turn trajectory, the predictive OP-Follower reduces horizontal pixel error by 57.2% and distance error by 50.5% relative to the non-predictive baseline. Similar improvements are observed for the U-turn case, with reductions of 43.4% in pixel error and 42.4% in distance error. These results demonstrate that delay-aware prediction is not merely a performance enhancement, but a fundamental requirement for maintaining stable and accurate closed-loop person following under realistic visual perception latency.

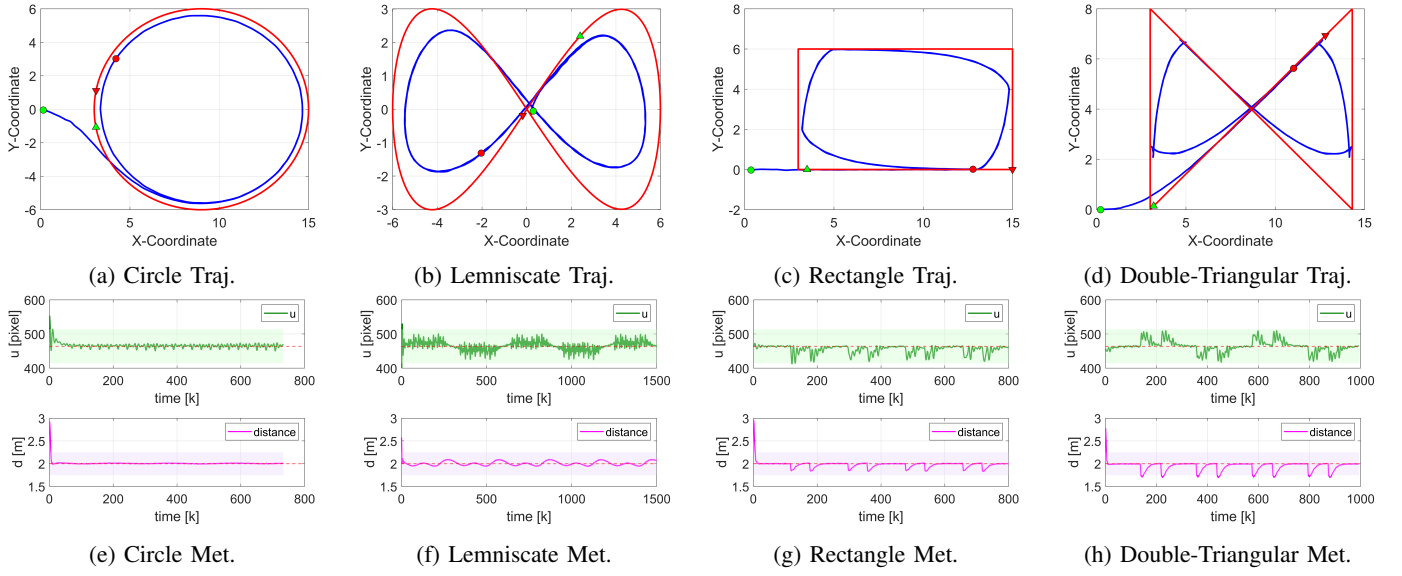


Fig. 5: **Simulation comparison of OP-Follower tracking performance under different trajectories.** Subfigures (a)-(d) depict the trajectories of the target person (red line) and the quadruped robot (blue line), with green and red markers denoting their respective start and end points. Subfigures (e)-(h) present the corresponding performance metrics, where the red dashed lines indicate the reference values: the image center at u -coordinate of 464, and the desired following distance of 2 m.

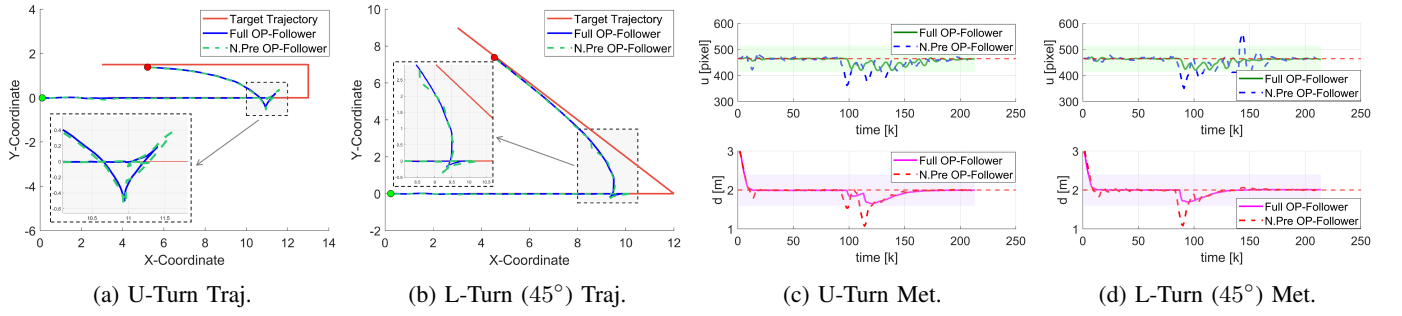


Fig. 6: **Comparison between predictive and N.Pre OP-Follower.** Subfigures (a)–(b) show the target and robot trajectories, while (c)–(d) report the image-space and distance tracking metrics.

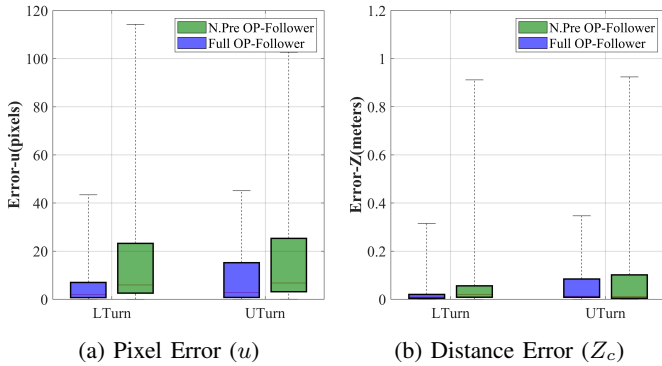
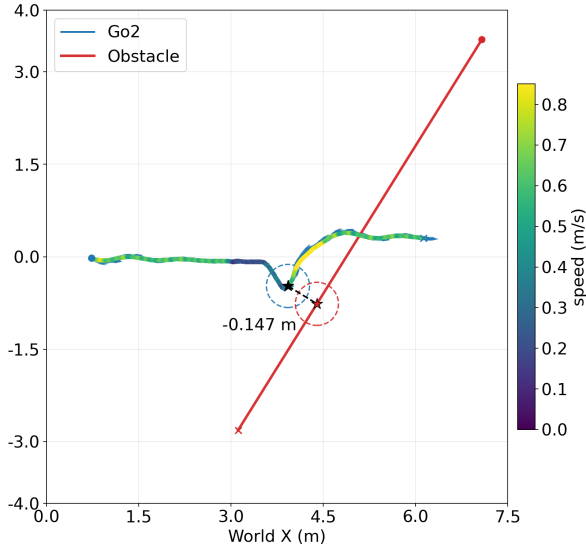


Fig. 7: **Box-plot comparison** of tracking errors for predictive and N.Pre OP-Follower.

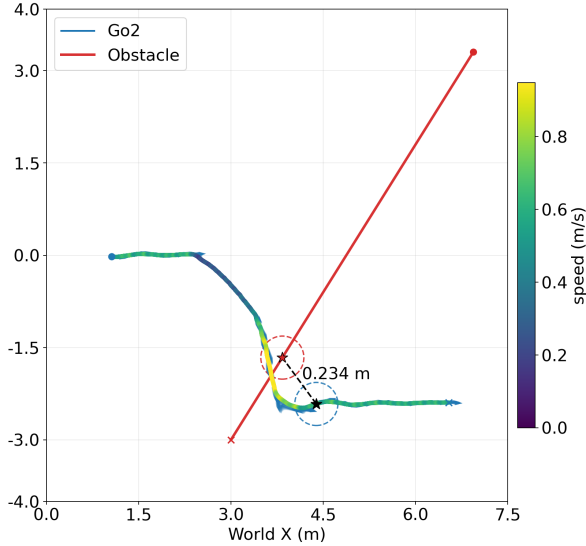
3) *Static APF vs. velocity-aware Dynamic APF:* To evaluate the contribution of the obstacle-avoidance formulation, we conducted a controlled module-level ablation comparing the

proposed velocity-aware Dynamic APF with a position-based Static APF baseline. We designed a dynamic crossing scenario in the high-fidelity Gazebo environment. The quadruped robot was commanded to maintain a nominal forward tracking velocity, while a dynamic obstacle rapidly cut across the robot's intended trajectory diagonally (from the front-left to the back-right). To isolate the algorithmic contribution, both configurations utilized the exact same visual perception pipeline, control bounds, and locomotion interface.

Under this controlled crossing scenario, the Static APF responded too late and resulted in a collision, whereas the velocity-aware formulation generated an earlier lateral correction and completed the maneuver with positive clearance. This result provides direct evidence that incorporating approach velocity improves the response to a fast-crossing obstacle under the tested conditions.



(a) Static APF (Collision)



(b) Velocity-aware Dynamic APF (Success)

Fig. 8: Modular ablation on dynamic obstacle avoidance. (a) Static APF fails to react to the dynamic threat in time, resulting in a collision. (b) Velocity-aware Dynamic APF anticipates the collision and executes an early tangential detour.

C. Real-World Experiments

We conduct extensive real-world experiments in both indoor and outdoor environments, including offices, long corridors, basketball courts, and pedestrian pathways, as illustrated in Fig. 9. These scenarios are designed to evaluate the robustness and practical deployability of the proposed framework under realistic conditions, such as intermittent occlusions, illumination variations, and frequent human–robot interactions. Due to the inherent difficulty of obtaining accurate ground-truth measurements in unconstrained environments, the experiments are recorded and evaluated through synchronized video streams.

Following common practice in real-world robot following studies, we assess system performance based on qualitative consistency, absence of persistent target loss, and sustained satisfaction of tracking and safety constraints over extended time horizons.

1) **Visual Obstacle Avoidance:** In this scenario, the robot follows a walking target while navigating around chairs, poles, pedestrians, and other irregular obstacles. When an obstacle enters the influence radius ρ_0 , the APF module generates a lateral corrective motion, while the IBVS–NMPC controller simultaneously adjusts the yaw rate to preserve visual alignment with the target. This decoupled strategy allows the robot to avoid obstacles at close range without inducing large deviations from the desired following trajectory or causing target loss.

2) **Multi-Person Following:** This experiment involves multiple people walking at different speeds and occasionally crossing paths in the robot’s field of view. The YOLO-World and BoxMOT pipeline maintains a persistent identity for the designated target person, while non-target individuals are automatically treated as dynamic obstacles. The robot demonstrates stable tracking behavior even under short-term occlusions and target reappearance, indicating reliable ID association and effective integration of perception and control.

3) **Dynamic Pursuit–Evasion Interaction:** In this challenging test, the target performs abrupt turns, accelerations, and evasive maneuvers. As shown in Fig. 11, the combination of KF–based prediction and the moving-horizon NMPC formulation enables the robot to maintain stable behavior even during temporary target disappearance. By continuing to generate control commands based on predicted target states, the system avoids persistent target loss and oscillatory responses. Once visual measurements are recovered, the robot rapidly and smoothly re-centers the target within its field of view.

4) **Long-Distance Outdoor Following:** We further evaluate long-range following performance in large outdoor environments over trajectories exceeding 600 m, including straight-line segments and curved paths around trees and street objects. The qualitative field performance snapshots from both the third-person and robot onboard perspectives are compiled in Fig. 10. As demonstrated by the visual sequences, the proposed OP-Follower maintains continuous target tracking with stable control inputs and consistent regulation of the socially compliant following distance, even under varying ambient lighting and environmental clutter.

To complement the qualitative snapshots with route-level evidence, Fig. 12 presents the GPS trajectory logs recorded using Mapviz. The logs cover two representative outdoor trials: a campus route A exceeding 600 m and a campus route B of 1.4 km run. The plotted trajectories contain local offsets relative to the underlying walkway geometry, particularly near dense tree canopies and buildings. These deviations are consistent with GNSS multipath effects and signal attenuation [34] and should therefore be interpreted as localization-logging uncertainty rather than direct tracking error. The recorded 1.4 km run was completed continuously without observed

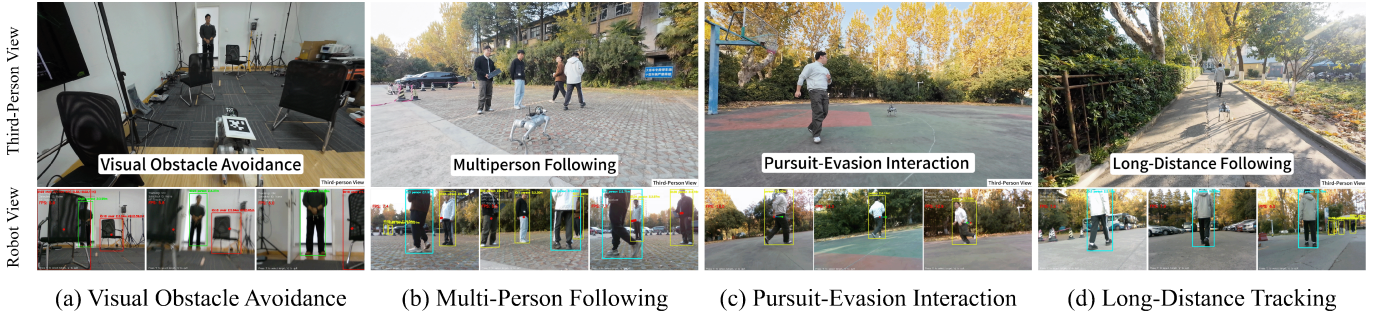


Fig. 9: **Movie: Real-world qualitative results of OP-Follower.** Representative indoor and outdoor scenarios used to validate robustness under dynamic conditions, including occlusions, lighting variations, and dense human activity.

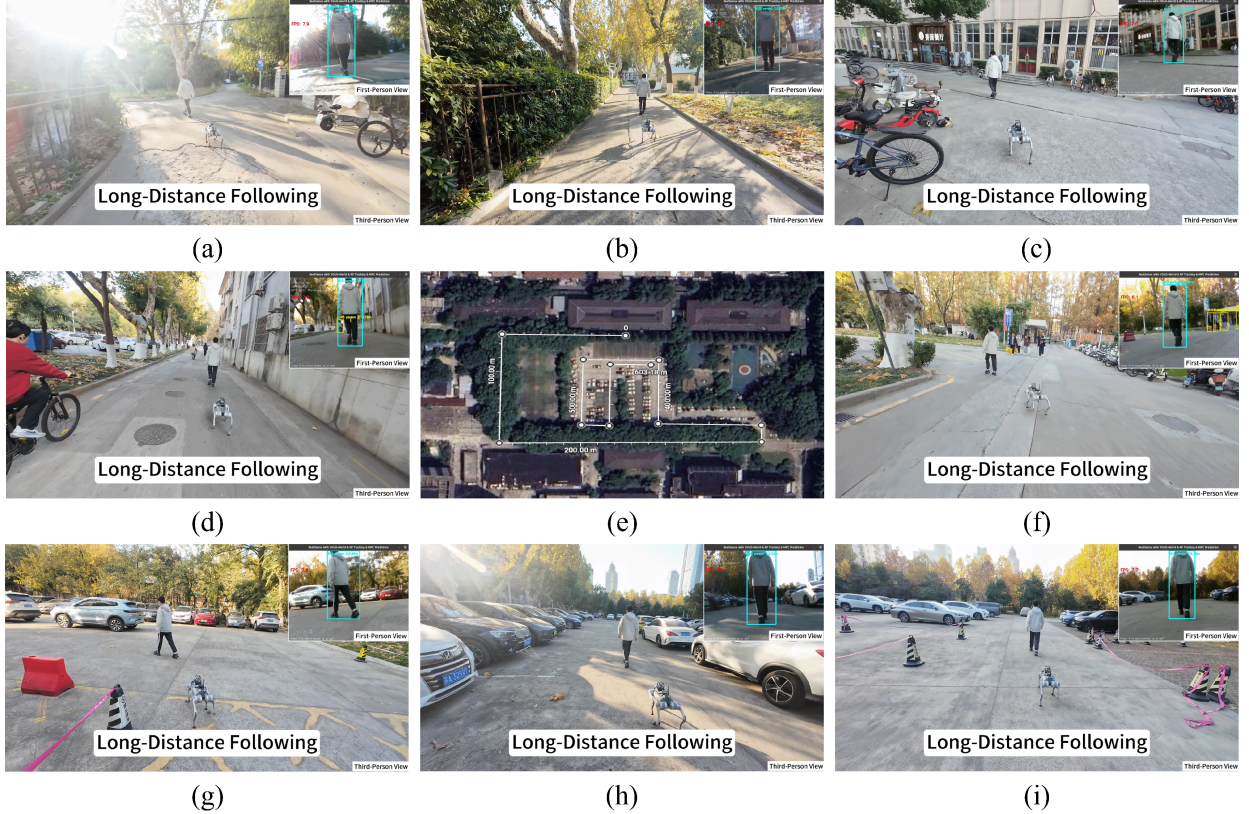


Fig. 10: **Movie: Real-world long-distance outdoor following performance.** Subfigure (e) shows a sketch map of the route, while subfigures (a)–(d) and (f)–(i) present snapshots of the robot during the long-distance following experiment.

manual takeover, providing additional evidence of the feasibility of long-horizon outdoor following.

5) **Quantitative Performance Analysis:** We further conduct a quantitative analysis of the proposed OP-Follower system based on onboard telemetry. It should be explicitly clarified that while metrics such as absolute target-loss rates, precise obstacle-clearance margins, and repeated-trial success statistics—benchmarked against external ground truth—are vital for a full-scale assessment, their rigorous derivation is inherently difficult in the context of our large-scale, unconstrained outdoor experiments. The absence of expansive external motion-capture infrastructure in these dynamic field environments necessitates a reliance on real-time visual feature estimates

and onboard logs. Within these methodological constraints, we prioritize the evaluation of tracking precision and consistency. Specifically, we introduce the concept of an Optimal Tracking Zone (OTZ) as a high-fidelity proxy to quantify the fine-grained regulation quality and reliability of the integrated perception-control loop under realistic deployment conditions. The boundaries of this zone are selected with reference to camera-centered regulation and HRI proxemics.

Fig. 13 illustrates the time-series trajectories of the horizontal image coordinate u and the relative distance Z_c . The results demonstrate that the horizontal tracking error is consistently regulated around the image center ($u_{ref}^c = 320$), and the depth



Fig. 11: **Target loss and recovery during pursuit-evasion interaction.** The target’s predicted velocity is visualized by the RGB trajectory lines from the robot’s perspective.



(a) Original >600 m tracking loop (b) Extended 1.4 km stress-test

Fig. 12: Mapviz-recorded GPS trajectories of the quadruped robot during long-distance field deployments.

measurement smoothly tracks the desired reference ($Z_{ref} = 2.0$ m). The system exhibits distinct dynamic responsiveness and stability, robustly maintaining the target lock even when the pedestrian performs sudden turns or rapid maneuvers.

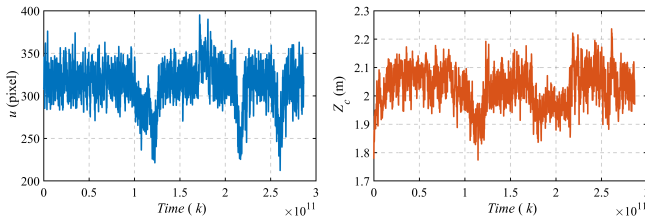


Fig. 13: **Quantitative results of the field experiment** based on onboard logs (Data source: reproduction of the Long-Distance Following experiment). (a) Variation of the horizontal pixel coordinate u of the pedestrian. (b) Relative distance Z_c between the quadruped robot and the pedestrian.

Furthermore, to provide a deeper evaluation of the system’s reliability, we compute comprehensive statistical metrics including the Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Maximum Absolute Error, as summarized in Table III. Traditional metrics such as “time

within bounds” often approach 100% due to the strict physical Field-of-View (FOV), offering limited insight into the fine-grained precision of the controller. Therefore, we introduce the concept of an Optimal Tracking Zone to evaluate steady-state performance.

For image-space regulation, we define a tight evaluation band of ± 50 pixels around the image center. For a 640×480 image, this band corresponds to 15.6% of the image width and is substantially narrower than the central 80% field-of-view constraint imposed by the controller. It therefore provides a stringent measure of horizontal centering performance. This threshold is used as an evaluation criterion rather than as a hard optical or safety boundary.

For depth regulation, the reference distance and its bounded tolerance are inspired by Hall’s classical proxemics theory [35]. Hall maps human spatial interactions into defined paradigms, notably the *Social-Consultative Zone* (1.2 m to 2.0 m) and the *Public Zone* (2.0 m to 3.6 m). We deliberately anchor the reference following distance Z_{ref} at 2.0 m—the transition boundary between these zones—to preserve psychological safety without degrading tracking responsiveness. The depth OTZ is subsequently defined as a tight ± 0.15 m margin around this reference. This tolerance restricts fluctuations to merely 18.75% of the Social Zone’s width and 9.375% of the Public Zone’s width.

As shown in Table III, the quadruped robot successfully maintained the target within these optimal, high-fidelity zones for 93.23% and 96.85% of the experimental duration, respectively. These metrics provide supporting evidence for the robustness and tracking accuracy of the integrated perception-control framework under real-world conditions.

TABLE III: Quantitative real-world analysis based on onboard logs for a representative run.

Metric	Value
<i>Image-space Error ($u - u_{ref}^c$)</i>	
MAE (pixels)	20.11
RMSE (pixels)	26.71
Max Absolute Error (pixels)	107.5
<i>Distance Error ($Z_c - Z_{ref}$)</i>	
MAE (meters)	0.0616
RMSE (meters)	0.0744
Max Absolute Error (meters)	0.2362
<i>Reliability & Consistency</i>	
Time within Image OTZ (%)	93.23%
Time within Distance OTZ (%)	96.85%

Note: These metrics are log-based onboard estimates rather than externally measured ground truth.

6) Computation Analysis: To quantify the computational load and real-time reliability of the OP-Follower on the resource-constrained Jetson Orin NX, Table IV summarizes the comprehensive execution time statistics—including Mean, Maximum, Minimum, and Variance—for each core module, measured directly from real-world onboard logs.

As expected, the perception front-end (YOLO-World detection and BoxMOT tracking) dominates the computational

cost. The statistics reveal that these visual modules exhibit high variance and substantial worst-case latency spikes (with maximums reaching 145.93 ms and 315.87 ms). These computational spikes typically occur during complex scenarios, such as feature re-identification after target occlusion or handling multiple tracked instances in cluttered backgrounds. In contrast, the closed-loop modules critical to motion smoothness and safety (KF prediction, APF avoidance, and IBVS-NMPC) are generally lightweight. As shown in Table IV, the variance for both the APF and MPC modules is near-zero (0.05 and 0.06, respectively). Even in the worst-case scenarios, the APF avoidance requires only 6.14 ms, and the ACADOS-based IBVS-NMPC solver remains at 5.19 ms per iteration.

These statistics explicitly validate the necessity of the proposed asynchronous perception-control pipeline. If the 50 Hz high-level controller relied strictly on synchronous visual updates, the observed perception latency spikes would inevitably violate the real-time constraint (20 ms budget). However, by employing a lightweight delay-aware Kalman Filter (Mean: 0.57 ms) to continuously propagate state estimates at the control rate, the framework effectively bridges the perception-control bandwidth mismatch. Consequently, the mean total computation of the high-level control loop stays well within the 20 ms budget, leaving sufficient margin for system overhead and guaranteeing stable real-time onboard execution even when visual inference lags.

It is noteworthy that while the statistical variance of the local control modules remains bounded, the maximum execution time of the KF occasionally exhibits latency spikes (e.g., up to 32.96 ms), which exceeds the nominal 20 ms budget of the 50 Hz control loop. These rare anomalies in long-duration field deployments are primarily attributed to OS-level thread scheduling jitter and occasional message queue contention, rather than algorithmic complexity. To strictly guarantee the 50 Hz command transmission, an asynchronous timeout mechanism is integrated into the control thread. If the KF state estimation is not refreshed within the strict 20 ms deadline, the control loop invokes a pure predictive forward-propagation using the most recent valid posterior state. This non-blocking architecture isolates the real-time NMPC solver from intermittent state-estimation jitters, ensuring the continuous and smooth 50 Hz execution.

TABLE IV: Computational load and real-time performance of OP-Follower on Jetson Orin NX.

Module	Mean (ms)	Max (ms)	Min (ms)	Var (ms ²)
YOLO-World	60.03	145.93	38.49	458.75
BoxMOT	47.22	315.87	1.20	780.50
APF	0.12	6.14	0.02	0.05
KF	0.57	32.96	0.09	3.88
MPC	1.20	5.19	0.71	0.06

D. Discussion

The experimental results demonstrate that OP-Follower achieves stable person following across a wide range of simulated and real-world scenarios using only onboard vision.

It is worth emphasizing the distinction between the system-level integration of OP-Follower and its methodological contributions. The perception front-end relies on SOTA vision models (e.g., YOLO-World and BoxMOT), which are not our standalone algorithmic novelties. Rather, the core contribution lies in the unified architecture that explicitly bridges the gap between low-frequency, delay-prone visual perception and high-rate, safety-aware control. Directly feeding low-rate bounding boxes to an NMPC or APF module inevitably leads to oscillatory or unsafe behaviors. By introducing an asynchronous state propagation mechanism combined with a decoupled control interface, our framework transforms off-the-shelf, intermittent visual detections into a stable, high-frequency control loop suitable for highly dynamic quadruped locomotion. Beyond the reported quantitative metrics, several key insights emerge regarding the system design choices and their implications for vision-driven quadruped navigation.

First, the results highlight the necessity of **delay-aware prediction** for vision-based person following on quadruped robots. Comparative simulations show that directly relying on low-frequency visual measurements leads to oscillatory behavior and degraded tracking performance, particularly during sharp turns or abrupt target motion. By integrating KF-based prediction with NMPC, OP-Follower compensates for perception latency and enables anticipatory control actions. This confirms that trajectory prediction is not merely a performance enhancement, but a fundamental enabler for stable closed-loop control under realistic onboard perception constraints, consistent with existing studies [8].

Second, the proposed **decoupled obstacle avoidance** strategy effectively resolves conflicts between visual servoing and collision avoidance. Instead of embedding obstacle constraints directly into the IBVS objective, which would significantly increase optimization complexity, OP-Follower separates longitudinal tracking and lateral avoidance commands. This design allows the robot to maintain visual alignment with the target while safely maneuvering around obstacles. No obvious conflicts were observed in our tested cluttered scenarios, which indicates such a decoupling provides a practical balance between control simplicity and behavioral robustness, particularly for vision-driven systems without explicit global maps.

Third, real-world experiments validate the feasibility of a **fully onboard, pure vision-based RPF system** in diverse and dynamic environments. The system maintains reliable target tracking under lighting variations, intermittent occlusions, multi-person interactions, and long-distance outdoor deployment, while operating entirely on a Jetson Orin NX. This observation aligns with recent work showing that modern visual perception, when combined with predictive control, can support real-time robot navigation without reliance on auxiliary sensors or offboard computation [4].

Importantly, this work does not aim to replace global planning or map-based navigation frameworks. Instead, OP-Follower focuses on enabling robust, fully onboard local autonomy for vision-based person following, where global maps, external infrastructure, or prior environment knowledge

are unavailable or undesirable. In this sense, the proposed framework is complementary to higher-level planning modules and can be naturally integrated into hierarchical navigation systems when global information is available.

Finally, limitations and potential failure modes of the proposed framework should be acknowledged. First, the vision-based perception pipeline may degrade under **prolonged or severe occlusions, dense crowds with frequent identity (ID) switches, or large illumination variations**, potentially leading to temporary target loss [13]. Second, the current KF prediction module relies on a local constant-velocity assumption; **abrupt target maneuvers** that strongly violate this assumption, especially when coupled with **strong body motion or camera disturbance** during highly dynamic quadruped locomotion, can temporarily degrade tracking accuracy and induce oscillatory commands. Third, the APF-based obstacle avoidance relies solely on locally perceived depth and does not guarantee global optimality. Although the decoupled tracking and avoidance architecture effectively mitigates classical collinear local minima, the purely reactive nature of APF can still lead to **lateral trapping or oscillatory behaviors in highly constrained spaces** (e.g., narrow corridors or U-shaped dead ends) [31]. Fourth, the reduced visual-state representation omits the vertical image coordinate. This simplification neglects the coupling among vertical image motion, camera attitude, and terrain elevation. On uneven terrain or steep slopes, the reduced state may exhibit increased model mismatch, degrading state prediction and closed-loop regulation. Extending the framework to uneven terrain would require a **full image-state or 3D relative-pose representation**, together with IMU-based camera-attitude compensation and terrain-elevation modeling. These limitations point to promising directions for future work, including the integration of stronger re-identification networks, more expressive and intent-aware prediction models [36], and tighter integration with higher-level global planning frameworks for uneven terrain.

VI. CONCLUSIONS

This paper presented **OP-Follower**, a fully onboard, pure vision-based framework for real-time, obstacle-aware person following on quadruped robots. By integrating open-vocabulary person detection, delay-aware predictive visual tracking, an IBVS–NMPC control architecture, and decoupled visual obstacle avoidance, the proposed system explicitly addresses the challenge of bridging low-frequency, delay-prone visual perception with high-rate, safety-aware control under onboard computational constraints. Extensive simulations and real-world experiments demonstrate that OP-Follower achieves robust and stable person following across diverse scenarios, including sharp turns, dynamic pursuit–evasion interactions, multi-person environments, and long-distance outdoor deployment, while operating entirely onboard without reliance on auxiliary sensors or offboard computation.

Future work will focus on improving person intent understanding, robust re-identification in crowded scenes, and

more expressive prediction models for handling highly dynamic interactions. Integrating semantic or terrain-aware perception may further enhance safety and adaptability, thereby extending OP-Follower toward more socially aware, context-aware, and long-horizon autonomous behaviors in real-world human–robot interaction.

REFERENCES

- [1] N. Hu, D. Su, S. Wang, X. Wang, H. Zhong, Z. Wang, Y. Qiao, and Y. Tan, “Segmentation and tracking of vegetable plants by exploiting vegetable shape feature for precision spray of agricultural robots,” *Journal of Field Robotics*, vol. 41, no. 3, pp. 570–586, 2024.
- [2] Z. Ma, Y. Lu, C. Huang, S. Feng, and J. Chen, “Research on automatic docking system of lng marine loading arm based on machine vision,” *Journal of Field Robotics*, vol. 42, no. 7, pp. 3213–3226, 2025.
- [3] H. Oleynikova, C. Lanegger, Z. Taylor, M. Pantic, A. Millane, R. Siegwart, and J. Nieto, “An open-source system for vision-based micro-aerial vehicle mapping, planning, and flight in cluttered environments,” *Journal of Field Robotics*, vol. 37, no. 4, pp. 642–666, 2020.
- [4] A. Saviolo, P. Rao, V. Radhakrishnan, J. Xiao, and G. Loianno, “Unifying foundation models with quadrotor control for visual tracking beyond object categories,” in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 7389–7396.
- [5] C. Scheidemann, L. Werner, V. Reijgwart, A. Cramariuc, J. Chomarat, J.-R. Chiu, R. Siegwart, and M. Hutter, “Obstacle-avoidant leader following with a quadruped robot,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2025, pp. 1407–1413.
- [6] C. Van Dang, H. Ahn, J.-W. Kim, and S. C. Lee, “Collision-free navigation in human-following task using a cognitive robotic system on differential drive vehicles,” *IEEE Transactions on Cognitive and Developmental Systems*, vol. 15, no. 1, pp. 78–87, 2022.
- [7] M. Gupta, S. Kumar, L. Behera, and V. K. Subramanian, “A novel vision-based tracking algorithm for a human-following mobile robot,” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 47, no. 7, pp. 1415–1427, 2016.
- [8] A. Bajcsy, A. Loquercio, A. Kumar, and J. Malik, “Learning vision-based pursuit-evasion robot policies,” in *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024, pp. 9197–9204.
- [9] Y. Zhan, H. Ye, and H. Zhang, “Monocular person localization under camera ego-motion,” in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2025, pp. 18 466–18 473.
- [10] J. Peng, Z. Liao, H. Yao, Z. Su, Y. Zeng, and H. Dai, “Mpc-based human-accompanying control strategy for improving the motion coordination between the target person and the robot,” in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2023, pp. 7969–7975.
- [11] J. Zhang, A. Li, Y. Qi, M. Li, J. Liu, S. Wang, H. Liu, G. Zhou, Y. Wu, X. Li *et al.*, “Embodied navigation foundation model,” *arXiv preprint arXiv:2509.12129*, 2025.
- [12] H. Ye, J. Zhao, Y. Pan, W. Cherr, L. He, and H. Zhang, “Robot person following under partial occlusion,” in *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2023, pp. 7591–7597.
- [13] H. Ye, J. Zhao, Y. Zhan, W. Chen, L. He, and H. Zhang, “Person re-identification for robot person following with online continual learning,” *IEEE Robotics and Automation Letters*, vol. 9, no. 11, pp. 9151–9158, 2024.
- [14] L. Zhao, J. Su, Y. Zhong, W. Xie, J. Su, X. Chen, C. Chen, and S. Li, “Beltlinenet: A shape-prior-guided lightweight network for real-time deviation detection in circular pipe conveyors,” *IEEE Sensors Journal*, vol. 25, no. 11, pp. 20 024–20 036, 2025.
- [15] W. Li, J. Su, C. Liu, W.-H. Chen, and S. Li, “Dr-mpc: Disturbance-resilient model predictive visual servoing control for quadrotor uav pipeline inspect,” in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2025, pp. 20 738–20 745.
- [16] H. Zhang, J. Ge, J. Su, K. Gu, F. Wang, W.-H. Chen, and S. Li, “Model predictive control with residual learning and real-time disturbance rejection: Design and experimentation,” *Control Engineering Practice*, vol. 165, p. 106587, 2025.

- [17] H. Zhang, J. Ge, J. Su, Y. Pan, J. Yang, W.-H. Chen, and S. Li, "Pred-mppi: Disturbance-preview and efficient mppi for robust quadrotor tracking with hardware validation," in *2026 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2026.
- [18] J. Su, B. Li, and W.-H. Chen, "On existence, optimality and asymptotic stability of the kalman filter with partially observed inputs," *Automatica*, vol. 53, pp. 149–154, 2015.
- [19] T. Cheng, L. Song, Y. Ge, W. Liu, X. Wang, and Y. Shan, "Yolo-world: Real-time open-vocabulary object detection," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2024, pp. 16 901–16 911.
- [20] M. Broström, "Boxmot: Pluggable sota multi-object tracking modules for segmentation, object detection and pose estimation models," 2026. [Online]. Available: <https://github.com/mikel-brostrom/boxmot>
- [21] J. Montesdeoca, J. M. Toibero, J. Jordan, A. Zell, and R. Carelli, "Person-following controller with socially acceptable robot motion," *Robotics and Autonomous Systems*, vol. 153, p. 104075, 2022.
- [22] F. Chaumette and S. Hutchinson, "Visual servo control. i. basic approaches," *IEEE Robotics & Automation Magazine*, vol. 13, no. 4, pp. 82–90, 2006.
- [23] G. Allibert, E. Courtial, and F. Chaumette, "Predictive control for constrained image-based visual servoing," *IEEE Transactions on Robotics*, vol. 26, no. 5, pp. 933–939, 2010.
- [24] S. Heshmati-Alamdari, G. C. Karras, A. Eqtami, and K. J. Kyriakopoulos, "A robust self triggered image based visual servoing model predictive control scheme for small autonomous robots," in *2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2015, pp. 5492–5497.
- [25] S. Leisiazar, S. R. R. Rohani, E. J. Park, A. Lim, and M. Chen, "Adapting to frequent human direction changes in autonomous frontal following robots," *IEEE Robotics and Automation Letters*, vol. 10, no. 3, pp. 2934–2941, 2025.
- [26] S. Wang, J. Zhang, M. Li, J. Liu, A. Li, K. Wu, F. Zhong, J. Yu, Z. Zhang, and H. Wang, "Trackvla: Embodied visual tracking in the wild," in *Conference on Robot Learning*. PMLR, 2025, pp. 4139–4164.
- [27] L. Shao, Z. Guo, J. Yang, and S. Li, "Vision-based quadrotor rapid landing control with an uncooperative platform: an alternating predictive observer approach," *IEEE Transactions on Intelligent Vehicles*, vol. 10, no. 4, pp. 2665–2678, 2024.
- [28] S. Poddar, C. Mavrogiannis, and S. S. Srinivasa, "From crowd motion prediction to robot navigation in crowds," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2023, pp. 6765–6772.
- [29] C. Schöller, V. Aravatinos, F. Lay, and A. Knoll, "What the constant velocity model can teach us about pedestrian motion prediction," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 1696–1703, 2020.
- [30] K. Mohta, M. Watterson, Y. Mulgaonkar, S. Liu, C. Qu, A. Makineni, K. Saulnier, K. Sun, A. Zhu, J. Delmerico *et al.*, "Fast, autonomous flight in gps-denied and cluttered environments," *Journal of Field Robotics*, vol. 35, no. 1, pp. 101–120, 2018.
- [31] S. Karaman and E. Frazzoli, "Sampling-based algorithms for optimal motion planning," *The international journal of robotics research*, vol. 30, no. 7, pp. 846–894, 2011.
- [32] P. Fiorini and Z. Shiller, "Motion planning in dynamic environments using velocity obstacles," *The international journal of robotics research*, vol. 17, no. 7, pp. 760–772, 1998.
- [33] R. Verschueren, G. Frison, D. Kouzoupis, J. Frey, N. v. Duijkeren, A. Zanelli, B. Novoselnik, T. Albin, R. Quirynen, and M. Diehl, "acados—a modular open-source framework for fast embedded optimal control," *Mathematical Programming Computation*, vol. 14, no. 1, pp. 147–183, 2022.
- [34] D. Maier and A. Kleiner, "Improved gps sensor model for mobile robots in urban terrain," in *2010 IEEE international conference on robotics and automation*. IEEE, 2010, pp. 4385–4390.
- [35] E. T. Hall and E. T. Hall, *The hidden dimension*. Anchor, 1966, vol. 609.
- [36] W. Situ, H. Ye, J. Peng, Y. Zhan, and H. Zhang, "Adap-rpf: Adaptive trajectory sampling for robot person following in dynamic crowded environments," *arXiv preprint arXiv:2510.11308*, 2025.